

Dionex_COMPASS_August2022

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Daily Set up

Read in and Format the raw data - change wd & file names

```
# SULFATE DATA:
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Sdat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202208_S04.txt", sep='\t' , header=T, skip=3)
head(Sdat)
```

```
##      X      X.1      X.2 IC.S04 IC.S04.1 IC.S04.2 IC.S04.3 IC.S04.4
## 1 1 Lab Blank Unknown   n.a.    n.a.    n.a.    n.a.    n.a.
## 2 2 Lab Blank Unknown   n.a.    n.a.    n.a.    n.a.    n.a.
## 3 3 Lab Blank Unknown   n.a.    n.a.    n.a.    n.a.    n.a.
## 4 4 Lab Blank Unknown 4.297  0.0001  0.09  0.0001    0
## 5 5 Lab Blank Unknown   n.a.    n.a.    n.a.    n.a.    n.a.
## 6 6 Lab Blank Unknown   n.a.    n.a.    n.a.    n.a.    n.a.
```

```
## Only keep the columns that we need
```

```
Sdat <- Sdat[ ,c(2,5)] # dont need this here
head(Sdat)
```

```
##      X.1 IC.S04.1
## 1 Lab Blank   n.a.
## 2 Lab Blank   n.a.
## 3 Lab Blank   n.a.
## 4 Lab Blank 0.0001
## 5 Lab Blank   n.a.
## 6 Lab Blank   n.a.
```

```
## Name the columns correctly
```

```
colnames(Sdat) <- c( "Sample_ID", "S04_ppm")
Sdat$Sample_ID <- as.factor(Sdat$Sample_ID)
Sdat$S04_ppm <- as.numeric(Sdat$S04_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Sdat <- as.data.frame(Sdat)
head(Sdat)
```

```
##   Sample_ID S04_ppm
## 1 Lab Blank      NA
## 2 Lab Blank      NA
## 3 Lab Blank      NA
## 4 Lab Blank    1e-04
## 5 Lab Blank      NA
## 6 Lab Blank      NA
```

```
#Chloride data
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Cldat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202208_Cl.txt",sep='\t' , header=T, skip=3)
head(Cldat)
```

```
##   X      X.1      X.2 IC.Cl IC.Cl.1 IC.Cl.2 IC.Cl.3 IC.Cl.4
## 1 1 Lab Blank Unknown  n.a.   n.a.   n.a.   n.a.   n.a.
## 2 2 Lab Blank Unknown  n.a.   n.a.   n.a.   n.a.   n.a.
## 3 3 Lab Blank Unknown  n.a.   n.a.   n.a.   n.a.   n.a.
## 4 4 Lab Blank Unknown  3.22  0.0215  31.11  0.0287   0.1
## 5 5 Lab Blank Unknown  n.a.   n.a.   n.a.   n.a.   n.a.
## 6 6 Lab Blank Unknown  3.217  0.013  79.27  0.0174   0.1
```

```
## Only keep the columns that we need
```

```
Cldat <- Cldat[,c(2,5)]
head(Cldat)
```

```
##      X.1 IC.Cl.1
## 1 Lab Blank  n.a.
## 2 Lab Blank  n.a.
## 3 Lab Blank  n.a.
## 4 Lab Blank  0.0215
## 5 Lab Blank  n.a.
## 6 Lab Blank  0.013
```

```
## Name the columns correctly
```

```
colnames(Cldat) <- c( "Sample_ID", "Cl_ppm")
Cldat$Sample_ID <- as.factor(Cldat$Sample_ID)
Cldat$Cl_ppm <- as.numeric(Cldat$Cl_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Cldat <- as.data.frame(Cldat)
head(Cldat)
```

```
##   Sample_ID Cl_ppm
## 1 Lab Blank      NA
## 2 Lab Blank      NA
## 3 Lab Blank      NA
## 4 Lab Blank 0.0215
## 5 Lab Blank      NA
## 6 Lab Blank 0.0130
```

```
## Bring the data back together:
```

```
all_dat <- merge(Sdat, Cldat, by="Sample_ID")
```

```
head(all_dat)
```

```
##      Sample_ID      S04_ppm      Cl_ppm
## 1          2659.6013 14576.8593
## 2          2659.6013  3622.8327
## 3          2659.6013    0.0084
## 4          2659.6013  3759.8455
## 5          2659.6013         NA
## 6          556.7597 14576.8593
```

```
## Remove empty lines
```

```
all_dat <- all_dat[!(is.na(all_dat$Sample_ID) | all_dat$Sample_ID==""), ]
```

```
head(all_dat)
```

```
##              Sample_ID      S04_ppm      Cl_ppm
## 26      1_GCrew_202208_UP_LysA_20cm    2.4251 10.7228
## 27     10_GCrew_202208_TR_LysB_20cm    1.9019  6.6606
## 28 10_GCrew_202208_TR_LysB_20cm_dup    2.0380  6.5797
## 29 10_GCrew_202208_TR_LysB_20cm_spk 108.4972 24.4734
## 30     11_GCrew_202208_TR_LysB_45cm    2.7212  3.1153
## 31     12_GCrew_202208_TR_LysC_10cm    5.3369 27.0484
```

```
all_dat <- droplevels(all_dat[!all_dat$Sample_ID == 'Lab Blank',])
```

```
head(all_dat)
```

```
##              Sample_ID      S04_ppm      Cl_ppm
## 26      1_GCrew_202208_UP_LysA_20cm    2.4251 10.7228
## 27     10_GCrew_202208_TR_LysB_20cm    1.9019  6.6606
## 28 10_GCrew_202208_TR_LysB_20cm_dup    2.0380  6.5797
## 29 10_GCrew_202208_TR_LysB_20cm_spk 108.4972 24.4734
## 30     11_GCrew_202208_TR_LysB_45cm    2.7212  3.1153
## 31     12_GCrew_202208_TR_LysC_10cm    5.3369 27.0484
```

```
## Remove samples that were ran several times and seeme dto increase over time for S04
```

```
all_dat <- all_dat %>%
```

```
  filter(Sample_ID != "33_GWI_202208_UP_LysB_10cm")
```

Pull out standards - could do some checks here if we want

```
stds <- all_dat[grep("Standard", all_dat$Sample_ID),]
```

```
head(stds)
```

```
##      Sample_ID      S04_ppm      Cl_ppm
## 78 Standard 1    0.4842  5.0438
## 79 Standard 1    0.4842  5.0704
## 80 Standard 1    0.4842  5.0358
## 81 Standard 1    0.4842  5.0613
## 82 Standard 1    0.4842  5.0642
## 83 Standard 1    0.4875  5.0438
```

```
stds_chk_S <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(S04_ppm), sd = sd(S04_ppm))
stds_chk_S$cv <- (stds_chk_S$sd/stds_chk_S$mean)*100
stds_chk_S$flag <- ifelse(stds_chk_S$cv <2, 'YES', 'NO, rerun')
head(stds_chk_S)
```

```
## # A tibble: 5 x 5
##   Sample_ID    mean      sd    cv flag
##   <fct>      <dbl>   <dbl> <dbl> <chr>
## 1 Standard 1  0.486 0.00183 0.378 YES
## 2 Standard 2  0.940 0.00549 0.584 YES
## 3 Standard 3  1.94  0.00444 0.228 YES
## 4 Standard 4  9.94  0.0232  0.233 YES
## 5 Standard 5 20.0  0.0279  0.139 YES
```

```
stds_chk_Cl <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(Cl_ppm), sd = sd(Cl_ppm))
stds_chk_Cl$cv <- (stds_chk_Cl$sd/stds_chk_Cl$mean)*100
stds_chk_Cl$flag <- ifelse(stds_chk_Cl$cv <2, 'YES', 'NO, rerun')
head(stds_chk_Cl)
```

```
## # A tibble: 5 x 5
##   Sample_ID    mean      sd    cv flag
##   <fct>      <dbl>   <dbl> <dbl> <chr>
## 1 Standard 1   5.06 0.0134 0.264 YES
## 2 Standard 2  10.3 0.0482 0.470 YES
## 3 Standard 3  20.2 0.0366 0.182 YES
## 4 Standard 4  99.3 0.141  0.142 YES
## 5 Standard 5 200.  0.247  0.123 YES
```

Calculate mmol/L concentrations

```
#remove standards from sample dataframe
sampledat <- all_dat[!grepl("Standard", all_dat$Sample_ID),]
head(sampledat)
```

```
##           Sample_ID S04_ppm Cl_ppm
## 1 1_GCrew_202208_UP_LysA_20cm 2.4251 10.7228
## 2 10_GCrew_202208_TR_LysB_20cm 1.9019 6.6606
## 3 10_GCrew_202208_TR_LysB_20cm_dup 2.0380 6.5797
## 4 10_GCrew_202208_TR_LysB_20cm_spk 108.4972 24.4734
## 5 11_GCrew_202208_TR_LysB_45cm 2.7212 3.1153
## 6 12_GCrew_202208_TR_LysC_10cm 5.3369 27.0484
```

```
# Constants needed for calculations:
clmw <- 35.45 #molecular weight of Chloride: 35.45
smw <- 32.06 #molecular weight of sulfur: 32.06
```

```

# Convert ppm to mmol/L
sampledat$SO4_mM <- (sampledat$SO4_ppm / smw)
sampledat$Cl_mM <- (sampledat$Cl_ppm / clmw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
sampledat$salinity <- ((1.8070 * sampledat$Cl_ppm) + 0.026) / 1000

head(sampledat)

```

```

##           Sample_ID  SO4_ppm  Cl_ppm    SO4_mM    Cl_mM
## 1  1_GCrew_202208_UP_LysA_20cm    2.4251  10.7228  0.07564255  0.3024767
## 2  10_GCrew_202208_TR_LysB_20cm    1.9019   6.6606  0.05932314  0.1878872
## 3 10_GCrew_202208_TR_LysB_20cm_dup    2.0380   6.5797  0.06356831  0.1856051
## 4 10_GCrew_202208_TR_LysB_20cm_spk  108.4972  24.4734  3.38419214  0.6903639
## 5   11_GCrew_202208_TR_LysB_45cm    2.7212   3.1153  0.08487835  0.0878787
## 6   12_GCrew_202208_TR_LysC_10cm    5.3369  27.0484  0.16646600  0.7630014
##      salinity
## 1 0.019402100
## 2 0.012061704
## 3 0.011915518
## 4 0.044249434
## 5 0.005655347
## 6 0.048902459

```

Pull out dups and check with percent difference

```

#Show me the data that we have from the calculations
head(sampledat)

```

```

##           Sample_ID  SO4_ppm  Cl_ppm    SO4_mM    Cl_mM
## 1  1_GCrew_202208_UP_LysA_20cm    2.4251  10.7228  0.07564255  0.3024767
## 2  10_GCrew_202208_TR_LysB_20cm    1.9019   6.6606  0.05932314  0.1878872
## 3 10_GCrew_202208_TR_LysB_20cm_dup    2.0380   6.5797  0.06356831  0.1856051
## 4 10_GCrew_202208_TR_LysB_20cm_spk  108.4972  24.4734  3.38419214  0.6903639
## 5   11_GCrew_202208_TR_LysB_45cm    2.7212   3.1153  0.08487835  0.0878787
## 6   12_GCrew_202208_TR_LysC_10cm    5.3369  27.0484  0.16646600  0.7630014
##      salinity
## 1 0.019402100
## 2 0.012061704
## 3 0.011915518
## 4 0.044249434
## 5 0.005655347
## 6 0.048902459

```

```
#pull out any rows that have "dup" in the SampleID column
dups <- sampledats %>%
  filter(str_detect(Sample_ID, "dup")) #have to change this to match data
head(dups)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 10_GCrew_202208_TR_LysB_20cm_dup    2.0380    6.5797  0.06356831  0.1856051
## 2 20_GCrew_202208_WC_SipB_45cm_dup  405.2386  3967.4580 12.64000624 111.9170099
## 3 30_GWI_202208_UP_LysA_10cm_dup   508.7221  3659.4661 15.86781347 103.2289450
## 4 40_GWI_202208_TR_LysA_45cm_dup   735.8219  8149.1088 22.95140050 229.8761298
## 5 50_GWI_202208_WC_SipB_10cm_dup 1310.7931  9277.3775 40.88562383 261.7031735
## 6 60_GWI_202208_RHZ_TR_SF_2_dup   613.3640  4550.2601 19.13175296 128.3571255
##      salinity
## 1  0.01191552
## 2  7.16922261
## 3  6.61268124
## 4 14.72546560
## 5 16.76424714
## 6  8.22234600
```

```
#remove these from sample dataframe in a new dataframe
sampledat2 <- sampledats %>%
  filter(!str_detect(Sample_ID, "dup")) %>%
  filter(!str_detect(Sample_ID, "spk"))
head(sampledat2)
```

```
##           Sample_ID S04_ppm  Cl_ppm  S04_mM  Cl_mM  salinity
## 1 1_GCrew_202208_UP_LysA_20cm  2.4251 10.7228 0.07564255 0.3024767 0.019402100
## 2 10_GCrew_202208_TR_LysB_20cm  1.9019  6.6606 0.05932314 0.1878872 0.012061704
## 3 11_GCrew_202208_TR_LysB_45cm  2.7212  3.1153 0.08487835 0.0878787 0.005655347
## 4 12_GCrew_202208_TR_LysC_10cm  5.3369 27.0484 0.16646600 0.7630014 0.048902459
## 5 13_GCrew_202208_TR_LysC_20cm  2.5114 35.9906 0.07833437 1.0152496 0.065061014
## 6 14_GCrew_202208_TR_LysC_45cm 20.8843  4.8610 0.65141298 0.1371227 0.008809827
```

```
#remove the dup from these IDs so we will have duplicate sample names
dups$Sample_ID<-gsub("_dup", "", as.character(dups$Sample_ID))
dups <- dups[ , -c(2,3)]
colnames(dups) <- c('Sample_ID', 'S04_mM_dup', 'Cl_mM_dup', 'salinity_dup')
head(dups)
```

```
##           Sample_ID  S04_mM_dup  Cl_mM_dup  salinity_dup
## 1 10_GCrew_202208_TR_LysB_20cm  0.06356831  0.1856051  0.01191552
## 2 20_GCrew_202208_WC_SipB_45cm 12.64000624 111.9170099  7.16922261
## 3 30_GWI_202208_UP_LysA_10cm 15.86781347 103.2289450  6.61268124
## 4 40_GWI_202208_TR_LysA_45cm 22.95140050 229.8761298 14.72546560
## 5 50_GWI_202208_WC_SipB_10cm 40.88562383 261.7031735 16.76424714
## 6 60_GWI_202208_RHZ_TR_SF_2 19.13175296 128.3571255  8.22234600
```

```
#put it back together with the old data set and look for duplicates
QAdups <- merge(sampledat2, dups)
head(QAdups)
```

```
##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1 10_GCrew_202208_TR_LysB_20cm   1.9019   6.6606  0.05932314  0.1878872
## 2 20_GCrew_202208_WC_SipB_45cm 412.4507 4032.5780 12.86496257 113.7539633
## 3 30_GWI_202208_UP_LysA_10cm 486.9415 3506.9076 15.18844354  98.9254612
## 4 40_GWI_202208_TR_LysA_45cm 658.9266 7268.5465 20.55291953 205.0365726
## 5 50_GWI_202208_WC_SipB_10cm 1397.0580 9892.3338 43.57635683 279.0503188
## 6 60_GWI_202208_RHZ_TR_SF_2 600.6481 4449.2865 18.73512477 125.5087870
##   salinity  S04_mM_dup  Cl_mM_dup salinity_dup
## 1 0.0120617 0.06356831 0.1856051 0.01191552
## 2 7.2868944 12.64000624 111.9170099 7.16922261
## 3 6.3370080 15.86781347 103.2289450 6.61268124
## 4 13.1342895 22.95140050 229.8761298 14.72546560
## 5 17.8754732 40.88562383 261.7031735 16.76424714
## 6 8.0398867 19.13175296 128.3571255 8.22234600
```

```
QAdups$S04_dups_chk <- ((abs(QAdups$S04_mM-QAdups$S04_mM_dup))/((QAdups$S04_mM+QAdups$S04_mM_dup)/2))*100
QAdups$S04_dups_flag <- ifelse(QAdups$S04_dups_chk <10, 'YES', 'NO, rerun')
```

```
QAdups$Cl_dups_chk <- ((abs(QAdups$Cl_mM-QAdups$Cl_mM_dup))/((QAdups$Cl_mM+QAdups$Cl_mM_dup)/2))*100
QAdups$Cl_dups_flag <- ifelse(QAdups$Cl_dups_chk <10, 'YES', 'NO, rerun')
```

```
head(QAdups)
```

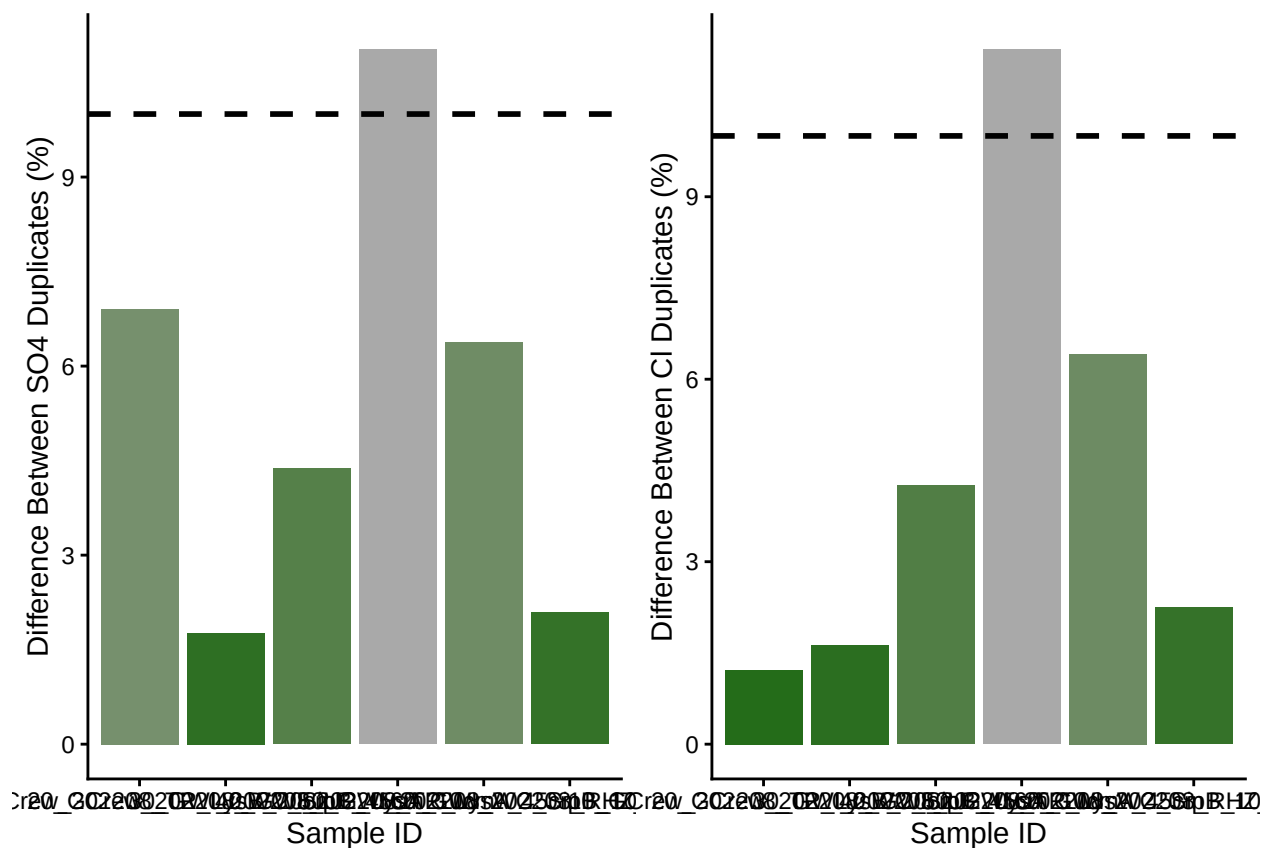
```
##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1 10_GCrew_202208_TR_LysB_20cm   1.9019   6.6606  0.05932314  0.1878872
## 2 20_GCrew_202208_WC_SipB_45cm 412.4507 4032.5780 12.86496257 113.7539633
## 3 30_GWI_202208_UP_LysA_10cm 486.9415 3506.9076 15.18844354  98.9254612
## 4 40_GWI_202208_TR_LysA_45cm 658.9266 7268.5465 20.55291953 205.0365726
## 5 50_GWI_202208_WC_SipB_10cm 1397.0580 9892.3338 43.57635683 279.0503188
## 6 60_GWI_202208_RHZ_TR_SF_2 600.6481 4449.2865 18.73512477 125.5087870
##   salinity  S04_mM_dup  Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag
## 1 0.0120617 0.06356831 0.1856051 0.01191552      6.908805      YES
## 2 7.2868944 12.64000624 111.9170099 7.16922261      1.764020      YES
## 3 6.3370080 15.86781347 103.2289450 6.61268124      4.375092      YES
## 4 13.1342895 22.95140050 229.8761298 14.72546560     11.026404 NO, rerun
## 5 17.8754732 40.88562383 261.7031735 16.76424714      6.371466      YES
## 6 8.0398867 19.13175296 128.3571255 8.22234600      2.094856      YES
##   Cl_dups_chk Cl_dups_flag
## 1      1.222027      YES
## 2      1.627993      YES
## 3      4.257621      YES
## 4     11.422778 NO, rerun
## 5      6.415916      YES
## 6      2.243971      YES
```

```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_chk, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
```

```
Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar, ncol=2, nrow=1)
```



```
Perc_dups <- QAdups %>%
  group_by(SO4_dups_flag) %>%
  summarise(S_no_rows = length(SO4_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(Cl_dups_flag) %>%
  summarise(Cl_no_rows = length(Cl_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "Cl_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)
```

```
Perc_dups2$Total <- length(QAdups$S04_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$Cl_Percent <- (Perc_dups2$Cl_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)
```

```
##          Flag S_no_rows      Flag Cl_no_rows Total S_Percent Cl_Percent
## 1 NO, rerun          1 NO, rerun          1      6 16.66667 16.66667
## 2      YES          5      YES          5      6 83.33333 83.33333
```

Pull out dups and check with cv

```
#the cv
# calculate the sd bewteen the two duplicates then divide by the mean and multiply by 100

df2 <- as.data.frame(QAdups$S04_mM)
df2$dups <- QAdups$S04_mM_dup

df2$sds <- apply(df2,1,sd)

QAdups$S04_dups_cv <- (df2$sds)/((QAdups$S04_mM+QAdups$S04_mM_dup)/2) * 100
QAdups$S04_dups_cv_flag <- ifelse(QAdups$S04_dups_cv <11, 'YES', 'NO, rerun')

head(QAdups)
```

```
##          Sample_ID      S04_ppm      Cl_ppm      S04_mM      Cl_mM
## 1 10_GCrew_202208_TR_LysB_20cm      1.9019      6.6606 0.05932314 0.1878872
## 2 20_GCrew_202208_WC_SipB_45cm 412.4507 4032.5780 12.86496257 113.7539633
## 3 30_GWI_202208_UP_LysA_10cm 486.9415 3506.9076 15.18844354 98.9254612
## 4 40_GWI_202208_TR_LysA_45cm 658.9266 7268.5465 20.55291953 205.0365726
## 5 50_GWI_202208_WC_SipB_10cm 1397.0580 9892.3338 43.57635683 279.0503188
## 6 60_GWI_202208_RHZ_TR_SF_2 600.6481 4449.2865 18.73512477 125.5087870
##          salinity S04_mM_dup      Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag
## 1 0.0120617 0.06356831 0.1856051 0.01191552 6.908805 YES
## 2 7.2868944 12.64000624 111.9170099 7.16922261 1.764020 YES
## 3 6.3370080 15.86781347 103.2289450 6.61268124 4.375092 YES
## 4 13.1342895 22.95140050 229.8761298 14.72546560 11.026404 NO, rerun
## 5 17.8754732 40.88562383 261.7031735 16.76424714 6.371466 YES
## 6 8.0398867 19.13175296 128.3571255 8.22234600 2.094856 YES
##          Cl_dups_chk Cl_dups_flag S04_dups_cv S04_dups_cv_flag
## 1 1.222027 YES 4.885263 YES
## 2 1.627993 YES 1.247350 YES
## 3 4.257621 YES 3.093657 YES
## 4 11.422778 NO, rerun 7.796845 YES
## 5 6.415916 YES 4.505306 YES
## 6 2.243971 YES 1.481287 YES
```

```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_cv, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
```

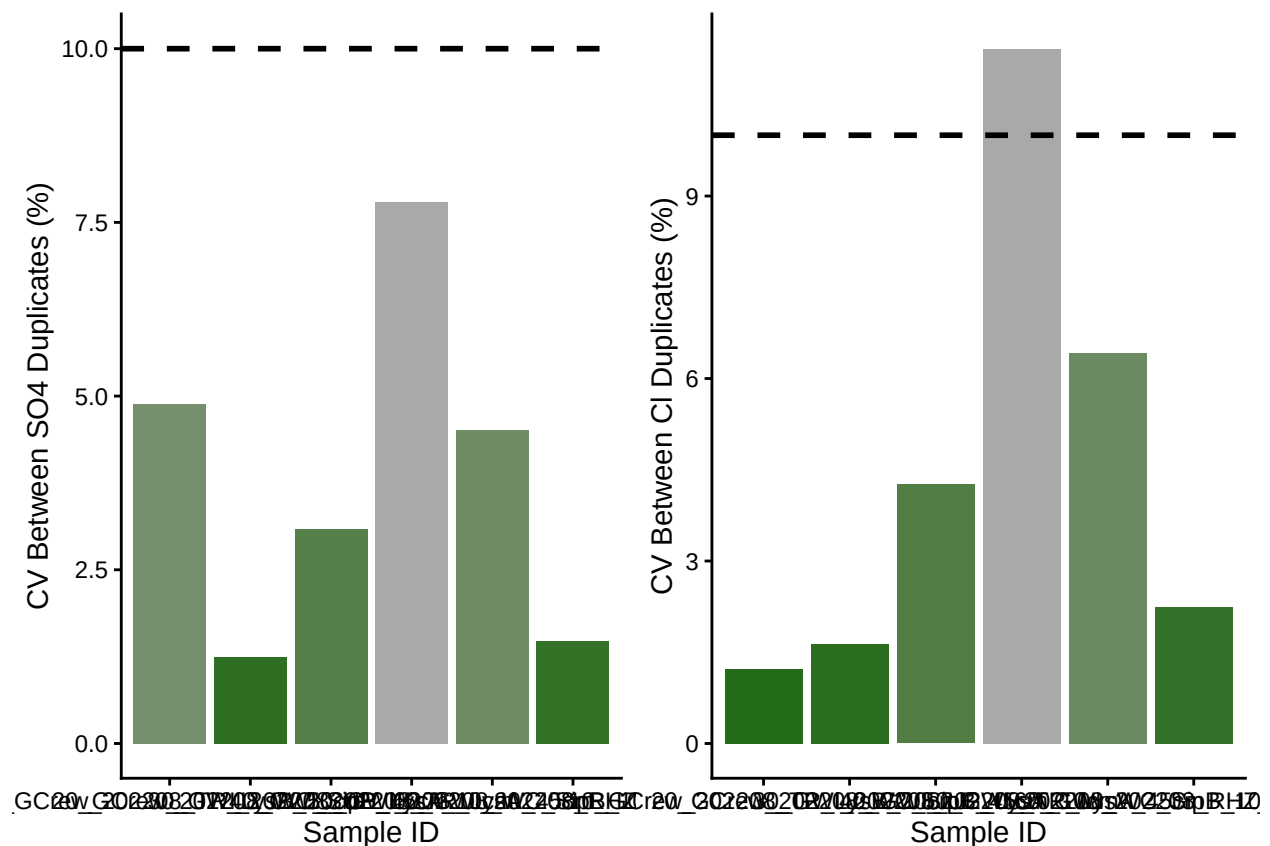
```

theme_classic() + labs(x= "Sample ID", y="CV Between S04 Duplicates (%)") +
theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
color = "black", size=1)

Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
geom_bar(stat = 'identity') +
scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
theme_classic() + labs(x= "Sample ID", y="CV Between Cl Duplicates (%)") +
theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar, ncol=2, nrow=1)

```



```

#check for percent of no, reruns to see if it would warrant reruns
Perc_dups <- QAdups %>%
  group_by(S04_dups_flag) %>%
  summarise(S_no_rows = length(S04_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(Cl_dups_flag) %>%
  summarise(Cl_no_rows = length(Cl_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "Cl_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)

```

```
Perc_dups2$Total <- length(QAdups$S04_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$Cl_Percent <- (Perc_dups2$Cl_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)
```

```
##          Flag S_no_rows      Flag Cl_no_rows Total S_Percent Cl_Percent
## 1 NO, rerun          1 NO, rerun          1      6 16.66667 16.66667
## 2      YES          5      YES          5      6 83.33333 83.33333
```

Pull out spikes and check

```
#Show me the data that we have from the calculations
head(sampledat)
```

```
##          Sample_ID S04_ppm Cl_ppm S04_mM Cl_mM
## 1 1_GCrew_202208_UP_LysA_20cm 2.4251 10.7228 0.07564255 0.3024767
## 2 10_GCrew_202208_TR_LysB_20cm 1.9019 6.6606 0.05932314 0.1878872
## 3 10_GCrew_202208_TR_LysB_20cm_dup 2.0380 6.5797 0.06356831 0.1856051
## 4 10_GCrew_202208_TR_LysB_20cm_spk 108.4972 24.4734 3.38419214 0.6903639
## 5 11_GCrew_202208_TR_LysB_45cm 2.7212 3.1153 0.08487835 0.0878787
## 6 12_GCrew_202208_TR_LysC_10cm 5.3369 27.0484 0.16646600 0.7630014
##          salinity
## 1 0.019402100
## 2 0.012061704
## 3 0.011915518
## 4 0.044249434
## 5 0.005655347
## 6 0.048902459
```

```
#pull out any rows that have "spk" in the SampleID column
spks <- sampledat %>%
  filter(str_detect(Sample_ID, "spk")) #have to change this to match data
head(spks)
```

```
##          Sample_ID S04_ppm Cl_ppm S04_mM Cl_mM
## 1 10_GCrew_202208_TR_LysB_20cm_spk 108.4972 24.4734 3.384192 0.6903639
## 2 20_GCrew_202208_WC_SipB_45cm_spk 620.2658 3954.0032 19.347031 111.5374669
## 3 30_GWI_202208_UP_LysA_10cm_spk 702.4759 3526.1084 21.911288 99.4670917
## 4 40_GWI_202208_TR_LysA_45cm_spk 855.0443 7143.8038 26.670128 201.5177377
## 5 50_GWI_202208_WC_SipB_10cm_spk 1854.8338 9964.9352 57.855078 281.0983131
## 6 60_GWI_202208_RHZ_TR_SF_2_spk 827.0714 4549.3304 25.797611 128.3308999
##          salinity
## 1 0.04424943
## 2 7.14490978
## 3 6.37170388
## 4 12.90887947
## 5 18.00666391
## 6 8.22066603
```

```
#remove the dup from these IDs so we will have duplicate sample names
spks$Sample_ID<-gsub("_spk","",as.character(spks$Sample_ID))
spks <- spks[ , -c(2,3, 5,6)]
colnames(spks) <- c('Sample_ID', 'S04_mM_spk')
head(spks)
```

```
##                Sample_ID S04_mM_spk
## 1 10_GCrew_202208_TR_LysB_20cm  3.384192
## 2 20_GCrew_202208_WC_SipB_45cm 19.347031
## 3 30_GWI_202208_UP_LysA_10cm 21.911288
## 4 40_GWI_202208_TR_LysA_45cm 26.670128
## 5 50_GWI_202208_WC_SipB_10cm 57.855078
## 6 60_GWI_202208_RHZ_TR_SF_2 25.797611
```

```
#put it back together with the old data set and look for duplicates
QAspks <- merge(sampledat, spks)
head(QAspks)
```

```
##                Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 10_GCrew_202208_TR_LysB_20cm  1.9019  6.6606  0.05932314  0.1878872
## 2 20_GCrew_202208_WC_SipB_45cm 412.4507 4032.5780 12.86496257 113.7539633
## 3 30_GWI_202208_UP_LysA_10cm 486.9415 3506.9076 15.18844354  98.9254612
## 4 40_GWI_202208_TR_LysA_45cm 658.9266 7268.5465 20.55291953 205.0365726
## 5 50_GWI_202208_WC_SipB_10cm 1397.0580 9892.3338 43.57635683 279.0503188
## 6 60_GWI_202208_RHZ_TR_SF_2 600.6481 4449.2865 18.73512477 125.5087870
##      salinity S04_mM_spk
## 1 0.0120617  3.384192
## 2 7.2868944 19.347031
## 3 6.3370080 21.911288
## 4 13.1342895 26.670128
## 5 17.8754732 57.855078
## 6 8.0398867 25.797611
```

```
#now we need to calculate the spike concentration and calculate the spike recovery
spkconc <- (250/smw)      # in mM
spkvol <- 10              # in uL
spkvol <- spkvol/1000000
#spike for these samples was 10uL of the 250mM standard
QAspks$S04_spk_Conc <- (spkconc)*spkvol      # mmoles of S04
head(QAspks)
```

```
##                Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 10_GCrew_202208_TR_LysB_20cm  1.9019  6.6606  0.05932314  0.1878872
## 2 20_GCrew_202208_WC_SipB_45cm 412.4507 4032.5780 12.86496257 113.7539633
## 3 30_GWI_202208_UP_LysA_10cm 486.9415 3506.9076 15.18844354  98.9254612
## 4 40_GWI_202208_TR_LysA_45cm 658.9266 7268.5465 20.55291953 205.0365726
## 5 50_GWI_202208_WC_SipB_10cm 1397.0580 9892.3338 43.57635683 279.0503188
## 6 60_GWI_202208_RHZ_TR_SF_2 600.6481 4449.2865 18.73512477 125.5087870
##      salinity S04_mM_spk S04_spk_Conc
## 1 0.0120617  3.384192 7.797879e-05
## 2 7.2868944 19.347031 7.797879e-05
## 3 6.3370080 21.911288 7.797879e-05
```

```
## 4 13.1342895 26.670128 7.797879e-05
## 5 17.8754732 57.855078 7.797879e-05
## 6 8.0398867 25.797611 7.797879e-05
```

```
#need to determine dilution factors and initial amount of sample added
#if your samples are all the same dilution just use the first line
#for Steph / COMPASS this depends on the site so...
QAspks$Dilution <- 1
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 200, QAspks$Dilution)
#QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$Dilution)
#QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$Dilution)
#QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$Dilution)

#Set Sample volumes in uL
QAspks$SampleVol <- 1
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 1462, QAspks$SampleVol)
#QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$SampleVol)
#QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$SampleVol)
#QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$SampleVol)

#change sample volume to L
QAspks$SampleVol <- QAspks$SampleVol/1000000
head(QAspks)
```

```
##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1 10_GCrew_202208_TR_LysB_20cm   1.9019   6.6606 0.05932314 0.1878872
## 2 20_GCrew_202208_WC_SipB_45cm 412.4507 4032.5780 12.86496257 113.7539633
## 3 30_GWI_202208_UP_LysA_10cm 486.9415 3506.9076 15.18844354 98.9254612
## 4 40_GWI_202208_TR_LysA_45cm 658.9266 7268.5465 20.55291953 205.0365726
## 5 50_GWI_202208_WC_SipB_10cm 1397.0580 9892.3338 43.57635683 279.0503188
## 6 60_GWI_202208_RHZ_TR_SF_2 600.6481 4449.2865 18.73512477 125.5087870
##      salinity S04_mM_spk S04_spk_Conc Dilution SampleVol
## 1 0.0120617 3.384192 7.797879e-05 1 1e-06
## 2 7.2868944 19.347031 7.797879e-05 1 1e-06
## 3 6.3370080 21.911288 7.797879e-05 1 1e-06
## 4 13.1342895 26.670128 7.797879e-05 1 1e-06
## 5 17.8754732 57.855078 7.797879e-05 1 1e-06
## 6 8.0398867 25.797611 7.797879e-05 1 1e-06
```

```

#gives us the total SO4 in the sample in mmoles
QAspks$SO4_Total_unspkd <- (QAspks$SO4_mM/QAspks$Dilution)*(QAspks$SampleVol)

##total SO4 in spiked sample in mmoles
QAspks$SO4_Total_spkd <- (QAspks$SO4_mM_spk/QAspks$Dilution)*(QAspks$SampleVol+spkvol)

QAspks$SO4_expctd_spkd <- (QAspks$SO4_Total_unspkd + QAspks$SO4_spk_Conc)
QAspks$spk_recovery <- (QAspks$SO4_Total_spkd/QAspks$SO4_expctd_spkd)*100
QAspks$SO4_spks_flag <- ifelse(QAspks$spk_recovery <=120 & QAspks$spk_recovery >=80 , 'YES', 'NO, rerun')

head(QAspks)

```

```

##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1 10_GCrew_202208_TR_LysB_20cm   1.9019   6.6606  0.05932314  0.1878872
## 2 20_GCrew_202208_WC_SipB_45cm 412.4507 4032.5780 12.86496257 113.7539633
## 3 30_GWI_202208_UP_LysA_10cm 486.9415 3506.9076 15.18844354  98.9254612
## 4 40_GWI_202208_TR_LysA_45cm 658.9266 7268.5465 20.55291953 205.0365726
## 5 50_GWI_202208_WC_SipB_10cm 1397.0580 9892.3338 43.57635683 279.0503188
## 6 60_GWI_202208_RHZ_TR_SF_2 600.6481 4449.2865 18.73512477 125.5087870
##      salinity S04_mM_spk S04_spk_Conc Dilution SampleVol S04_Total_unspkd
## 1 0.0120617  3.384192 7.797879e-05      1      1e-06  5.932314e-08
## 2 7.2868944 19.347031 7.797879e-05      1      1e-06  1.286496e-05
## 3 6.3370080 21.911288 7.797879e-05      1      1e-06  1.518844e-05
## 4 13.1342895 26.670128 7.797879e-05      1      1e-06  2.055292e-05
## 5 17.8754732 57.855078 7.797879e-05      1      1e-06  4.357636e-05
## 6 8.0398867 25.797611 7.797879e-05      1      1e-06  1.873512e-05
##      S04_Total_spkd S04_expctd_spkd spk_recovery S04_spks_flag
## 1 3.722611e-05 7.803811e-05 47.70248 NO, rerun
## 2 2.128173e-04 9.084375e-05 234.26744 NO, rerun
## 3 2.410242e-04 9.316723e-05 258.70058 NO, rerun
## 4 2.933714e-04 9.853171e-05 297.74314 NO, rerun
## 5 6.364059e-04 1.215551e-04 523.55320 NO, rerun
## 6 2.837737e-04 9.671391e-05 293.41561 NO, rerun

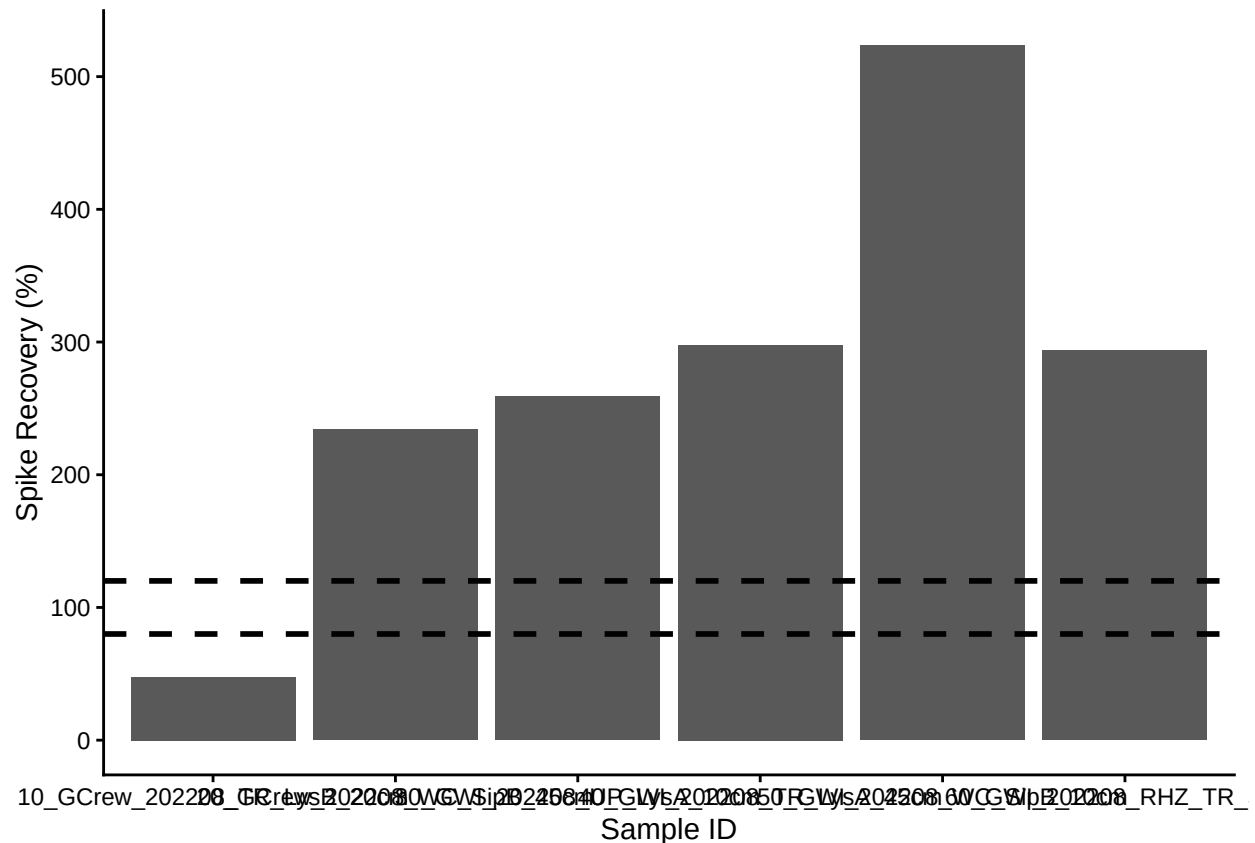
```

```

#plot spk recoveries output as a bar graph to easily check - want any over 10% to be red need to work on
spksbar <- ggplot(data = QAspks, aes(x = Sample_ID, y = spk_recovery)) +
  geom_bar(stat = 'identity') +
  scale_fill_manual(values=c("lightblue4")) +
  theme_classic() + labs(x= "Sample ID", y="Spike Recovery (%)") +
  theme(legend.position="none") +
  geom_hline(yintercept=80, linetype="dashed", color = "black", size=1) +
  geom_hline(yintercept=120, linetype="dashed", color = "black", size=1)

spksbar

```



```
#check for percent of no, reruns to see if it would warrant reruns
Perc_spks <- QAspks %>%
  group_by(S04_spks_flag) %>%
  summarise(no_rows = length(S04_spks_flag))
Perc_spks$Total <- length(QAspks$S04_spks_flag)
Perc_spks$Percent <- (Perc_spks$no_rows / Perc_spks$Total)*100
head(Perc_spks)
```

```
## # A tibble: 1 x 4
##   S04_spks_flag no_rows Total Percent
##   <chr>         <int> <int>   <dbl>
## 1 NO, rerun           6     6    100
```

Make final dataframe with IDs

```
#for steph <- pull out identifiers of the sample names
#pull the sample ID and separate it by the underscores
IDs <- data.frame(do.call('rbind', strsplit(as.character(sampled2$Sample_ID), '_ ', fixed=TRUE)))

## Warning in rbind(c("1", "GCrew", "202208", "UP", "LysA", "20cm"), c("10", :
## number of columns of result is not a multiple of vector length (arg 1)
```

```
colnames(IDs) <- c("Analysis_No" , "Site", "Date", "Zone", "Replicate", "Depth", "Tree")
head(IDs)
```

```
##   Analysis_No Site   Date Zone Replicate Depth Tree
## 1           1 GCrew 202208  UP      LysA   20cm    1
## 2          10 GCrew 202208  TR      LysB   20cm   10
## 3          11 GCrew 202208  TR      LysB   45cm   11
## 4          12 GCrew 202208  TR      LysC   10cm   12
## 5          13 GCrew 202208  TR      LysC   20cm   13
## 6          14 GCrew 202208  TR      LysC   45cm   14
```

```
#rejoin them to the dataframe
alldat <- cbind(IDs, sampled2)
head(alldat)
```

```
##   Analysis_No Site   Date Zone Replicate Depth Tree
## 1           1 GCrew 202208  UP      LysA   20cm    1
## 2          10 GCrew 202208  TR      LysB   20cm   10
## 3          11 GCrew 202208  TR      LysB   45cm   11
## 4          12 GCrew 202208  TR      LysC   10cm   12
## 5          13 GCrew 202208  TR      LysC   20cm   13
## 6          14 GCrew 202208  TR      LysC   45cm   14
##               Sample_ID S04_ppm Cl_ppm   S04_mM   Cl_mM   salinity
## 1 1_GCrew_202208_UP_LysA_20cm 2.4251 10.7228 0.07564255 0.3024767 0.019402100
## 2 10_GCrew_202208_TR_LysB_20cm 1.9019  6.6606 0.05932314 0.1878872 0.012061704
## 3 11_GCrew_202208_TR_LysB_45cm 2.7212  3.1153 0.08487835 0.0878787 0.005655347
## 4 12_GCrew_202208_TR_LysC_10cm 5.3369 27.0484 0.16646600 0.7630014 0.048902459
## 5 13_GCrew_202208_TR_LysC_20cm 2.5114 35.9906 0.07833437 1.0152496 0.065061014
## 6 14_GCrew_202208_TR_LysC_45cm 20.8843  4.8610 0.65141298 0.1371227 0.008809827
```

Make final dataframe with IDs

```
write.csv(alldat, file="Processed Data/COMPASS_SynopticCB_PW_Processed_Cl_S04_202208.csv")
```

#Change f

END